

Human Reproduction

2.0

Reproductive Events in Humans

As you are aware, **humans are sexually reproducing and viviparous**. Human reproduction runs through a fixed series of events, each with its own name.

1

Gametogenesis - the formation of gametes: **sperms** in males and **ovum** in females.

2

Insemination - the **transfer of sperms into the female genital tract**.

3

Fertilisation - the **fusion of male and female gametes**, leading to formation of the **zygote**.

4

Formation and development of the **blastocyst** and its **attachment to the uterine wall (implantation)**.

5

Gestation - the embryonic development inside the uterus.

6

Parturition - the delivery of the baby.

All these reproductive events **occur only after puberty**. There is an important difference between the sexes in how long gamete formation lasts: **sperm formation continues even in old men, but formation of ovum ceases in women around the age of fifty years**.

2.1

The Male Reproductive System

The **male reproductive system is located in the pelvis region**. It includes a **pair of testes** along with **accessory ducts, glands** and the **external genitalia**.

2.1a

The Testes

The **testes are situated outside the abdominal cavity** within a pouch called the **scrotum**. The scrotum **maintains the low temperature of the testes** - 2-2.5°C lower than the **normal internal body temperature** - which is necessary for spermatogenesis.

In adults, **each testis is oval**, with a **length of about 4 to 5 cm** and a **width of about 2 to 3 cm**. The testis is **covered by a dense covering**. Each testis has **about 250 compartments called testicular lobules**, and **each lobule contains one to three highly coiled seminiferous tubules** in which sperms are produced.

Each **seminiferous tubule** is lined on its inside by **two types of cells**: the **male germ cells (spermatogonia)** and the **Sertoli cells**. The **male germ cells undergo meiotic divisions**, finally leading to **sperm formation**, while the **Sertoli cells provide nutrition to the germ cells**.

The regions **outside the seminiferous tubules**, called **interstitial spaces**, contain **small blood vessels** and **interstitial cells**, also called **Leydig cells**. The **Leydig cells synthesise and secrete testicular hormones called androgens**. Other **immunologically competent cells** are also present.

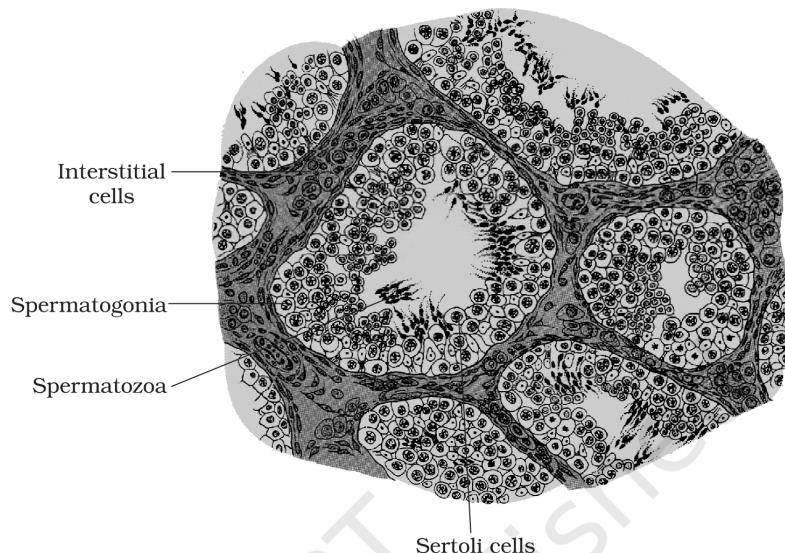


Fig. 2.2 - Diagrammatic sectional view of a seminiferous tubule. Labelled parts: interstitial cells, spermatogonia, spermatozoa and Sertoli cells.

The **male sex accessory ducts** include the **rete testis**, **vasa efferentia**, **epididymis** and **vas deferens**. Their connected path carries the sperms outward:

- 1 The **seminiferous tubules** of the testis open into the **vasa efferentia** through the **rete testis**.
- 2 The **vasa efferentia** leave the testis and open into the **epididymis**, located **along the posterior surface of each testis**.
- 3 The **epididymis** leads to the **vas deferens**, which **ascends to the abdomen and loops over the urinary bladder**.
- 4 The **vas deferens** receives a duct from the **seminal vesicle** and opens into the **urethra** as the **ejaculatory duct**.
- 5 The **urethra** originates from the **urinary bladder** and **extends through the penis** to its external opening called the **urethral meatus**.

Together, **these ducts store and transport the sperms** from the testis to the outside through the urethra.

The **penis is the male external genitalia**. It is made up of a **special tissue that helps in erection of the penis to facilitate insemination**. The **enlarged end of the penis**, called the **glans penis**, is covered by a loose fold of skin called the **foreskin**.

The **male accessory glands** include the **paired seminal vesicles**, a **prostate** and the **paired bulbourethral glands**. The **secretions of these glands constitute the seminal plasma**, which is **rich in fructose, calcium and certain enzymes**. The **secretions of the bulbourethral glands** also help in the **lubrication of the penis**.

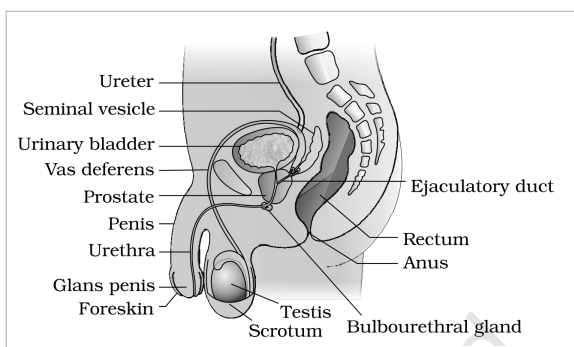


Fig. 2.1a - Diagrammatic sectional view of the male pelvis showing the reproductive system. Labelled parts: ureter, urinary bladder, seminal vesicle, ejaculatory duct, prostate, bulbourethral gland, vas deferens, urethra, penis, glans penis, foreskin, testis, scrotum, rectum and anus.

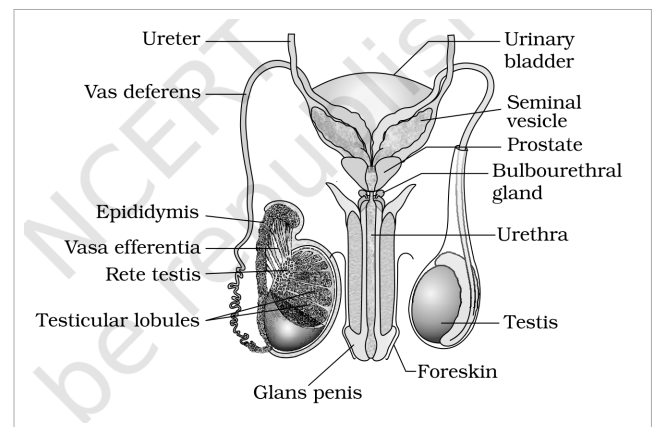


Fig. 2.1b - Diagrammatic view of the male reproductive system (part of the testis is opened to show inner details). Labelled parts: ureter, urinary bladder, seminal vesicle, prostate, bulbourethral gland, vas deferens, epididymis, vasa efferentia, rete testis, testicular lobules, urethra, glans penis, foreskin and testis.

The **female reproductive system** consists of a **pair of ovaries** along with a **pair of oviducts**, a **uterus**, a **cervix**, a **vagina** and the **external genitalia**, located in the **pelvic region**. These parts, along with a **pair of mammary glands**, are **integrated structurally and functionally** to support **ovulation, fertilisation, pregnancy, birth and child care**.

The **ovaries** are the **primary female sex organs** that **produce the female gamete (ovum)** and **several steroid hormones (ovarian hormones)**. They are **located one on each side of the lower abdomen**. Each ovary is about **2 to 4 cm in length** and is **connected to the pelvic wall and uterus by ligaments**.

Each ovary is **covered by a thin epithelium** which **encloses the ovarian stroma**. The **stroma is divided into two zones** - a **peripheral cortex** and an **inner medulla**. **Ovarian follicles in different stages of development are embedded in the stroma**.

The **oviducts (fallopian tubes)**, **uterus** and **vagina** constitute the **female accessory ducts**. Each fallopian tube is about **10-12 cm long** and **extends from the periphery of each ovary to the uterus**. Its parts, from the ovary towards the uterus, are:

- 1** **Infundibulum** - the **funnel-shaped part closer to the ovary**. Its edges have **finger-like projections called fimbriae**, which **help in the collection of the ovum after ovulation**.
- 2** **Ampulla** - the **wider part of the oviduct** that the infundibulum leads into.
- 3** **Isthmus** - the **last part of the oviduct**, which has a **narrow lumen** and **joins the uterus**.

The **uterus is single** and is also called the **womb**. Its **shape is like an inverted pear**, and it is **supported by ligaments attached to the pelvic wall**. The **uterus opens into the vagina through a narrow cervix**. The **cavity of the cervix is called the cervical canal**, which **along with the vagina forms the birth canal**.

The **wall of the uterus has three layers of tissue**:

Layer	Position	Nature / feature
Perimetrium	External	Thin membranous layer
Myometrium	Middle	Thick layer of smooth muscle; exhibits strong contraction during delivery of the baby
Endometrium	Inner	Glandular layer that lines the uterine cavity and undergoes cyclical changes during the menstrual cycle

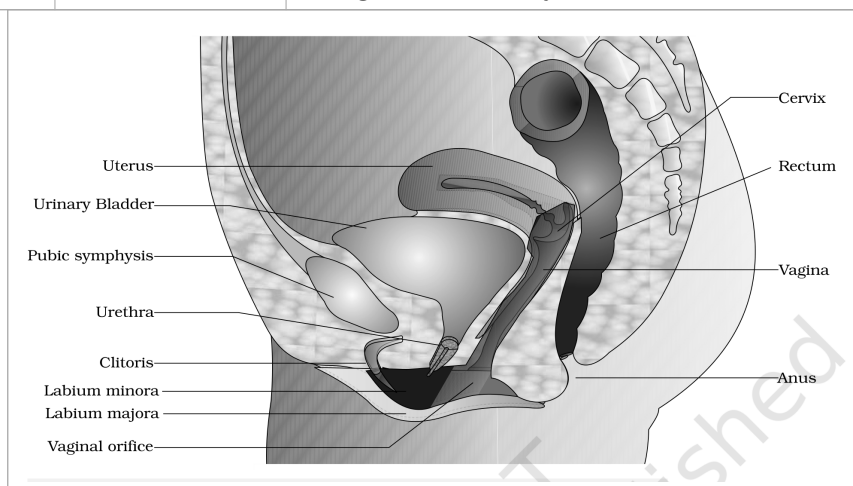


Fig. 2.3a - Diagrammatic sectional view of the female pelvis showing the reproductive system. Labelled parts: uterus, urinary bladder, pubic symphysis, urethra, clitoris, labium minora, labium majora, vaginal orifice, cervix, rectum, vagina and anus.

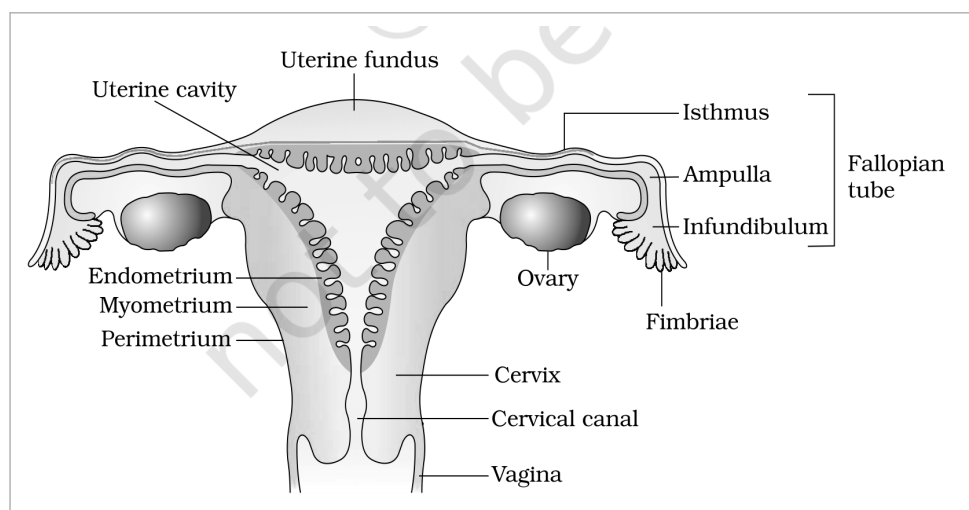


Fig. 2.3b - Diagrammatic sectional view of the female reproductive system. Labelled parts: uterine fundus, uterine cavity, endometrium, myometrium, perimetrium, isthmus, ampulla, infundibulum, fallopian tube, ovary, fimbriae, cervix, cervical canal and vagina.

The **female external genitalia** include the **mons pubis**, **labia majora**, **labia minora**, **hymen** and **clitoris**.

- **Mons pubis** - a cushion of fatty tissue covered by skin and pubic hair.
- **Labia majora** - fleshy folds of tissue which extend down from the mons pubis and surround the vaginal opening.
- **Labia minora** - paired folds of tissue under the labia majora.
- **Hymen** - a membrane that **often partially covers the opening of the vagina**.
- **Clitoris** - a **tiny finger-like structure** which lies at the upper junction of the two labia minora above the urethral opening.

① **[NOTE]** *The hymen is often torn during the first coitus (intercourse). It can also be broken by a sudden fall or jolt, insertion of a vaginal tampon, or active participation in some sports like horseback riding, cycling, etc. In some women the hymen persists even after coitus. Therefore the presence or absence of the hymen is not a reliable indicator of virginity or sexual experience.*

A functional mammary gland is characteristic of all female mammals, and the mammary glands are one of the female secondary sexual characteristics. The mammary glands are paired structures (breasts) that contain glandular tissue and a variable amount of fat. Milk drains from the gland along a fixed path:

- 1 The glandular tissue of each breast is divided into 15-20 mammary lobes containing clusters of cells called alveoli.
- 2 The cells of the alveoli secrete milk, which is stored in the cavities (lumens) of the alveoli.
- 3 The alveoli open into mammary tubules.
- 4 The tubules of each lobe join to form a mammary duct.
- 5 Several mammary ducts join to form a wider mammary ampulla, which is connected to a lactiferous duct and opens at the nipple, from where milk is expressed during breastfeeding.

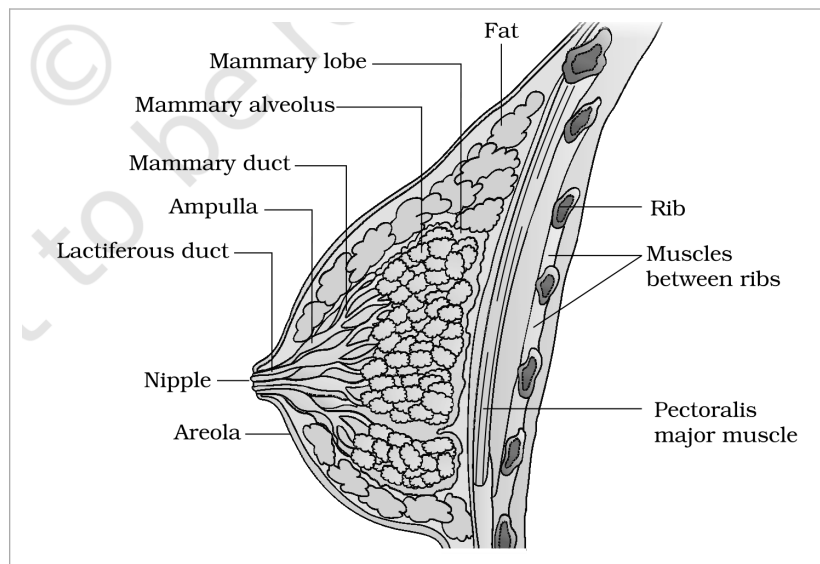


Fig. 2.4 - A diagrammatic sectional view of a mammary gland. Labelled parts: mammary lobe, mammary alveolus, mammary duct, ampulla, lactiferous duct, nipple, areola, fat, rib, muscles between ribs and pectoralis major muscle.

The **primary sex organs** - the **testis in males** and the **ovaries in females** - produce gametes, i.e. sperms and ovum respectively, by the process called **gametogenesis**.

In the testis, **immature male germ cells (spermatogonia)** produce sperms by spermatogenesis, a process that begins at puberty. The steps are:

1

Spermatogonia (sing. spermatogonium) on the inside wall of the seminiferous tubules **multiply by mitotic division and increase in number**. Each spermatogonium is **diploid and contains 46 chromosomes**.

2

Some spermatogonia, called **primary spermatocytes**, **periodically undergo meiosis**.

3

A **primary spermatocyte completes the first meiotic division (reduction division)**, producing **two equal, haploid secondary spermatocytes**, each with **only 23 chromosomes**.

4

The **secondary spermatocytes undergo the second meiotic division** to produce **four equal, haploid spermatids**.

5

The **spermatids are transformed into spermatozoa (sperms)** by the process called **spermiogenesis**.

6

After spermiogenesis, the **sperm heads become embedded in the Sertoli cells** and are **finally released from the seminiferous tubules** by the process called **spermiation**.

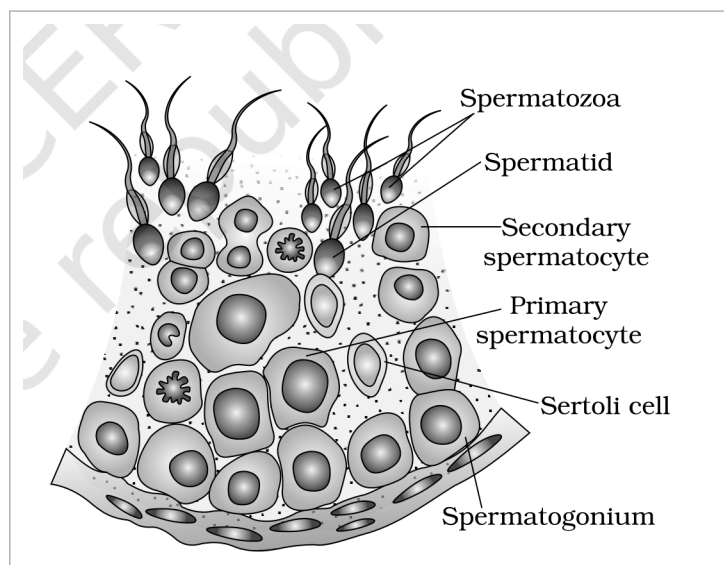


Fig. 2.5 - Diagrammatic sectional view of a seminiferous tubule (enlarged). Labelled cells: spermatogonium, primary spermatocyte, secondary spermatocyte, spermatid, spermatozoa and Sertoli cell.

2.3b

Hormonal Control of Spermatogenesis

Spermatogenesis **starts at puberty** due to a **significant increase in the secretion of gonadotropin releasing hormone (GnRH)**, which is a **hypothalamic hormone**.

1

Increased GnRH acts at the anterior pituitary gland and stimulates secretion of two gonadotropins - luteinising hormone (LH) and follicle stimulating hormone (FSH).

2

LH acts at the Leydig cells and stimulates synthesis and secretion of androgens. The androgens, in turn, stimulate the process of spermatogenesis.

3

FSH acts on the Sertoli cells and stimulates secretion of some factors which help in the process of spermiogenesis.

2.3c

Structure of the Sperm

A **sperm is a microscopic structure composed of a head, a neck, a middle piece and a tail**. A **plasma membrane envelops the whole body of the sperm**.

- **Head:** contains an **elongated haploid nucleus**, the **anterior portion of which is covered by a cap-like structure, the acrosome**. The acrosome is filled with **enzymes that help fertilisation of the ovum**.
- **Middle piece:** possesses **numerous mitochondria**, which **produce energy for the movement of the tail** that facilitates **sperm motility essential for fertilisation**.

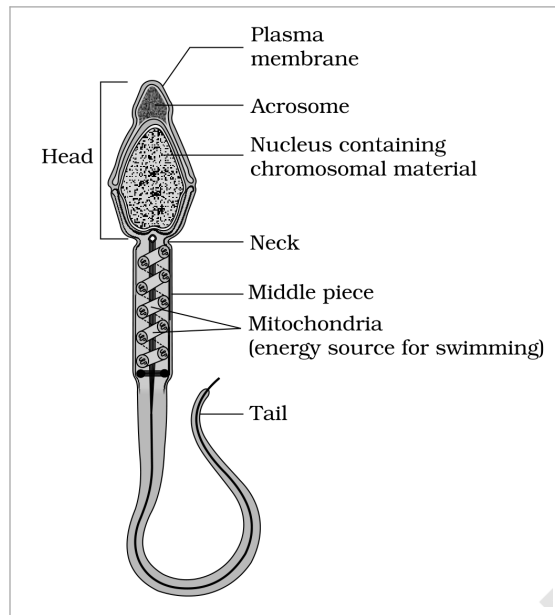


Fig. 2.6 - Structure of a sperm. Labelled parts: plasma membrane, acrosome, nucleus, head, neck, middle piece, mitochondria and tail.

The human male ejaculates about 200 to 300 million sperms during a coitus. For normal fertility, at least 60 per cent of sperms must have normal shape and size and at least 40 per cent must show vigorous motility.

Sperms released from the seminiferous tubules are transported by the accessory ducts. The secretions of the epididymis, vas deferens, seminal vesicle and prostate are essential for the maturation and motility of sperms. The seminal plasma along with the sperms constitutes the semen. The functions of the male sex accessory ducts and glands are maintained by the testicular hormones (androgens).

2.3d

Oogenesis

Oogenesis - the formation of a mature female gamete - is markedly different from spermatogenesis. It is initiated during the embryonic development stage, when a couple of million gamete mother cells (oogonia) are formed within each fetal ovary. No more oogonia are formed and added after birth.

- 1 Oogonia start division and enter prophase-I of the meiotic division, then get temporarily arrested at that stage; they are now called **primary oocytes**.
- 2 Each **primary oocyte** gets surrounded by a layer of **granulosa cells** and is called a **primary follicle**. A large number of these follicles degenerate during the phase from birth to puberty, so that at puberty only 60,000-80,000 primary follicles are left in each ovary.
- 3 The **primary follicles** get surrounded by more layers of **granulosa cells** and a new **theca** and are called **secondary follicles**.
- 4 A secondary follicle **transforms into a tertiary follicle**, characterised by a **fluid-filled cavity called the antrum**. The **theca layer** is organised into an inner **theca interna** and an outer **theca externa**.
- 5 At the **tertiary-follicle stage** the **primary oocyte** grows in size and completes its first meiotic division. This is an **unequal division**, resulting in a **large haploid secondary oocyte** and a **tiny first polar body**; the **secondary oocyte** retains the bulk of the nutrient-rich cytoplasm of the primary oocyte.
- 6 The **tertiary follicle** further changes into the **mature follicle or Graafian follicle**. The **secondary oocyte** forms a new membrane called the **zona pellucida** surrounding it.
- 7 The **Graafian follicle** ruptures to release the **secondary oocyte (ovum)** from the ovary by the process called **ovulation**.

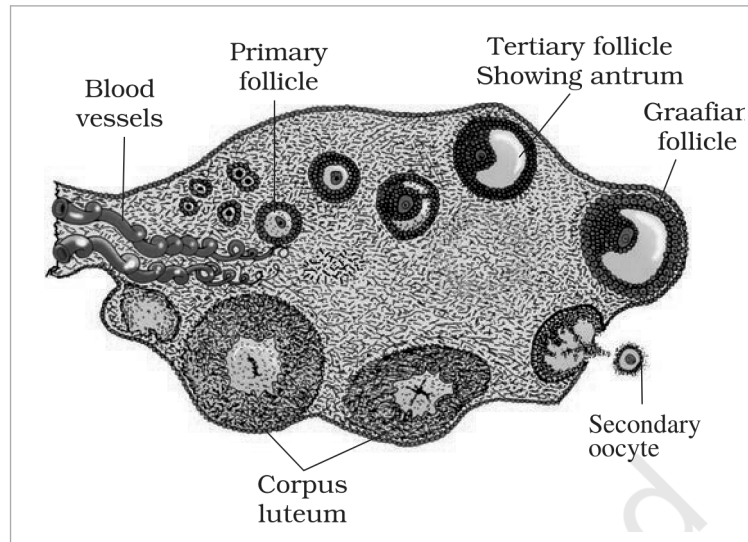


Fig. 2.7 - Diagrammatic sectional view of an ovary. Labelled parts: blood vessels, primary follicle, tertiary follicle, antrum, Graafian follicle, secondary oocyte and corpus luteum.

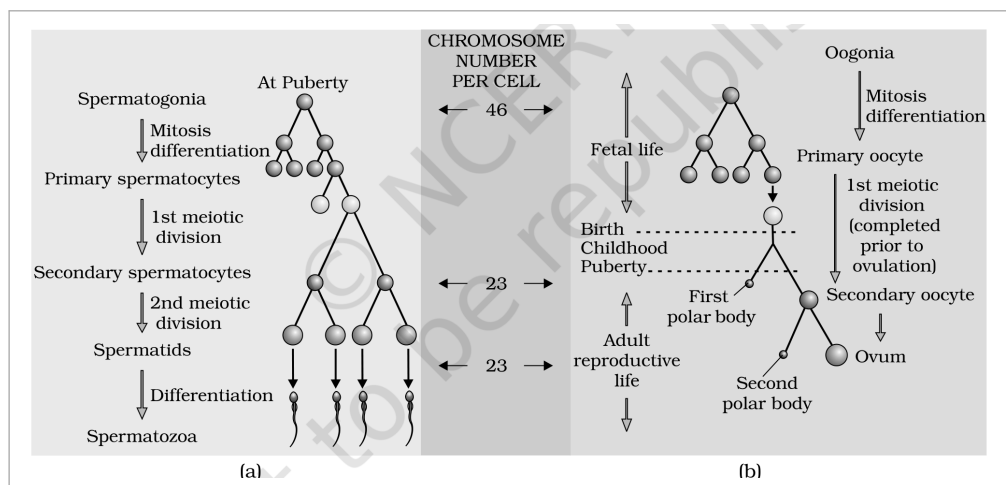


Fig. 2.8 - Schematic representation of (a) spermatogenesis and (b) oogenesis. Labelled: spermatogonia, mitosis, primary spermatocytes, secondary spermatocytes, spermatids and spermatozoa, with the chromosome number per cell shown as 46 and 23; and oogonia, primary oocyte, secondary oocyte, first polar body, second polar body and ovum.

2.4 Menstrual Cycle

The **reproductive cycle in the female primates** (e.g. monkeys, apes and human beings) is called the **menstrual cycle**. The **first menstruation begins at puberty** and is called **menarche**.

In human females, **menstruation is repeated at an average interval of about 28/29 days**. The **cycle of events starting from one menstruation till the next is called the menstrual cycle**, and **one ovum is released (ovulation) during the middle of each menstrual cycle**.

The cycle proceeds through the following phases:

- 1 **Menstrual phase:** the cycle starts with the menstrual phase, when menstrual flow occurs; it lasts for 3-5 days. The menstrual flow results due to breakdown of the endometrial lining of the uterus and its blood vessels, which forms a liquid that comes out through the vagina.
- 2 **Follicular phase:** the primary follicles grow to become a fully mature Graafian follicle and, simultaneously, the endometrium regenerates through proliferation.
- 3 **Ovulatory phase:** the LH surge induces rupture of the Graafian follicle and release of the ovum (ovulation).
- 4 **Luteal phase:** the remaining parts of the Graafian follicle transform into the corpus luteum, which secretes large amounts of progesterone.

① [NOTE] **Menstruation only occurs if the released ovum is not fertilised. A lack of menstruation may be indicative of pregnancy; it may also be caused due to some other underlying causes like stress, poor health, etc.**

These changes in the ovary and the uterus are induced by changes in the levels of pituitary and ovarian hormones. The secretion of gonadotropins (LH and FSH) increases gradually during the follicular phase and stimulates follicular development as well as the secretion of estrogens by the growing follicles. Both LH and FSH attain a peak level in the middle of the cycle (about the 14th day). The rapid secretion of LH leading to its maximum level during mid-cycle, called the LH surge, induces rupture of the Graafian follicle and release of the ovum (ovulation).

The progesterone secreted by the corpus luteum is essential for maintenance of the endometrium. Such an endometrium is necessary for implantation of the fertilised ovum and other events of pregnancy. During pregnancy all events of the menstrual cycle stop and there is no menstruation. In the absence of fertilisation, the corpus luteum degenerates, causing disintegration of the endometrium leading to menstruation, which marks a new cycle.

In human beings, menstrual cycles cease around 50 years of age; that is termed menopause. Cyclic menstruation is an indicator of a normal reproductive phase and extends between menarche and menopause.

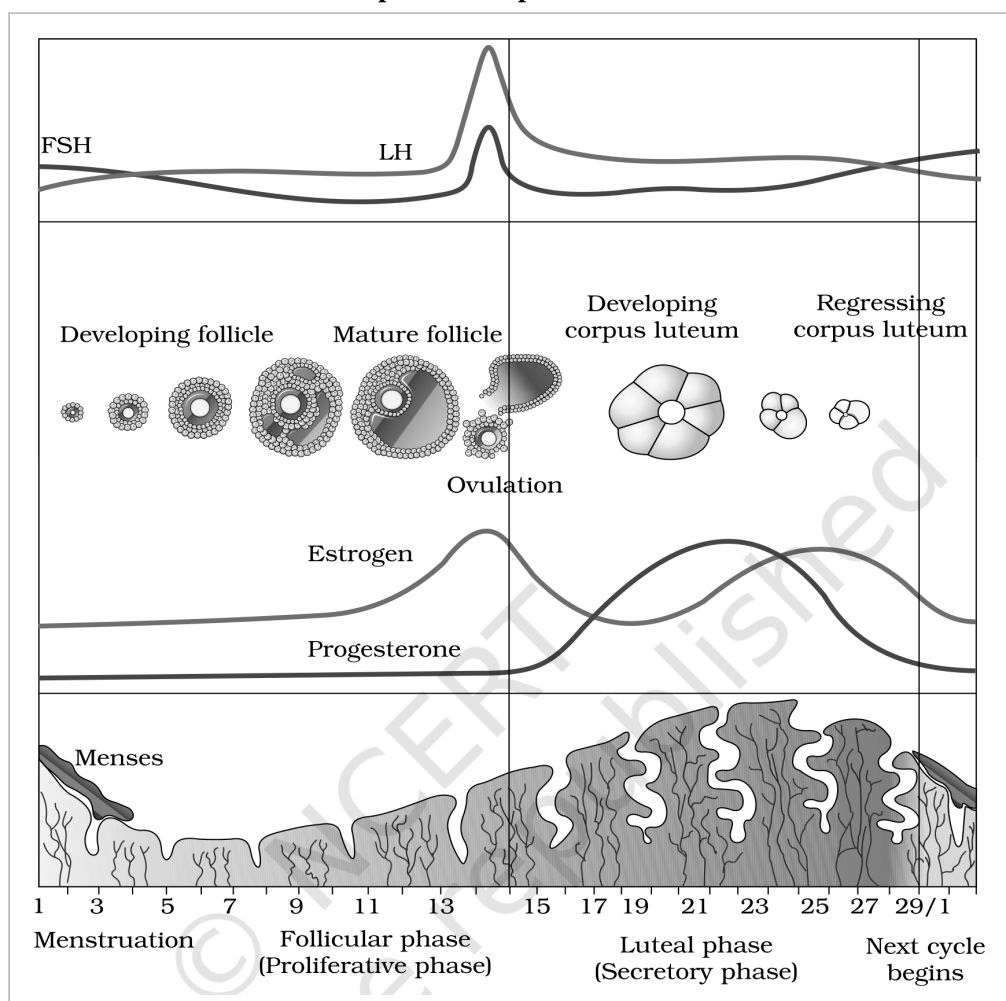


Fig. 2.9 - Diagrammatic presentation of the various events during a menstrual cycle. Labelled: FSH, LH, developing follicle, mature follicle, corpus luteum, ovulation, estrogen, progesterone, menses, the follicular phase and the luteal phase.

① [NOTE] **Menstrual Hygiene.** Maintenance of hygiene and sanitation during menstruation is very important. Take a bath and clean regularly; use **sanitary napkins** or **clean homemade pads**; change them **after every 4-5 hrs** as per requirement; dispose of used napkins by wrapping them in used paper; **do not throw used napkins in the drainpipe of toilets or in the open area**; and after handling the napkin, **wash hands with soap**.

During **copulation (coitus)**, semen is released by the penis into the vagina (insemination). The **motile sperms** swim rapidly, pass through the cervix, enter the uterus and finally reach the **ampullary region** of the fallopian tube. The **ovum** released by the ovary is also transported to the ampullary region, where **fertilisation** takes place.

① **[NOTE]** Fertilisation can only occur if the ovum and sperms are transported simultaneously to the ampullary region. This is why not all copulations lead to fertilisation and pregnancy.

The **process of fusion** of a sperm with an ovum is called **fertilisation**. During fertilisation, a **sperm** comes in contact with the **zona pellucida** layer of the ovum and induces changes in the membrane that block the entry of additional sperms. This ensures that only one sperm can fertilise an ovum. The secretions of the **acrosome** help the sperm enter the cytoplasm of the ovum through the **zona pellucida** and the **plasma membrane**.

This entry induces **completion of the meiotic division** of the **secondary oocyte**. The **second meiotic division** is also **unequal** and results in a **second polar body** and a **haploid ovum (ootid)**. The **haploid nucleus** of the sperm and that of the ovum fuse together to form a **diploid zygote**.

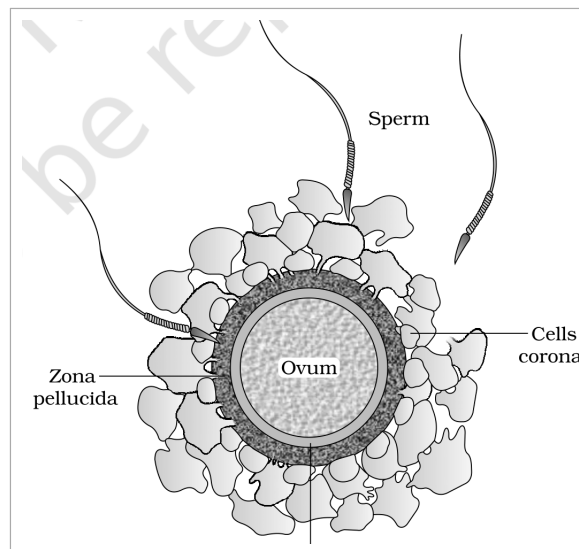


Fig. 2.10 - Ovum surrounded by a few sperms. Labelled parts: sperm, zona pellucida, corona radiata, perivitelline space and ovum.

The **sex of the baby** is decided at the **fertilisation stage**. The **chromosome pattern** in the human female is **XX** and in the male is **XY**. All **haploid gametes (ova)** produced by the female have the sex chromosome **X**, whereas in **male gametes (sperms)** the sex chromosome could be either **X** or **Y** - 50 per cent of sperms carry **X** and the other 50 per cent carry **Y**.

Thus the **zygote** carries either **XX** or **XY** depending on whether the **X-** or **Y-bearing sperm** fertilised the **ovum**: **XX** develops into a **female baby** and **XY** into a **male** (more on chromosomal patterns in Chapter 5). Therefore, **scientifically** it is correct to say that the **sex of the baby** is **determined by the father** and not by the mother.

1

Mitotic division called **cleavage** starts as the **zygote** moves through the **isthmus** of the **oviduct** towards the **uterus** and forms **2, 4, 8, 16** daughter cells called **blastomeres**.

2

The **embryo** with **8 to 16 blastomeres** is called a **morula**.

3

The **morula** continues to divide and transforms into a **blastocyst** as it moves further into the uterus. The **blastomeres** in the **blastocyst** are arranged into an **outer layer** called the **trophoblast** and an **inner group** of cells attached to the trophoblast called the **inner cell mass**.

4

The **trophoblast layer** gets attached to the **endometrium** and the **inner cell mass** gets differentiated as the **embryo**.

5

After attachment, the **uterine cells** divide rapidly and cover the **blastocyst**, so the **blastocyst** becomes **embedded in the endometrium** - this is called **implantation**, and it leads to **pregnancy**.

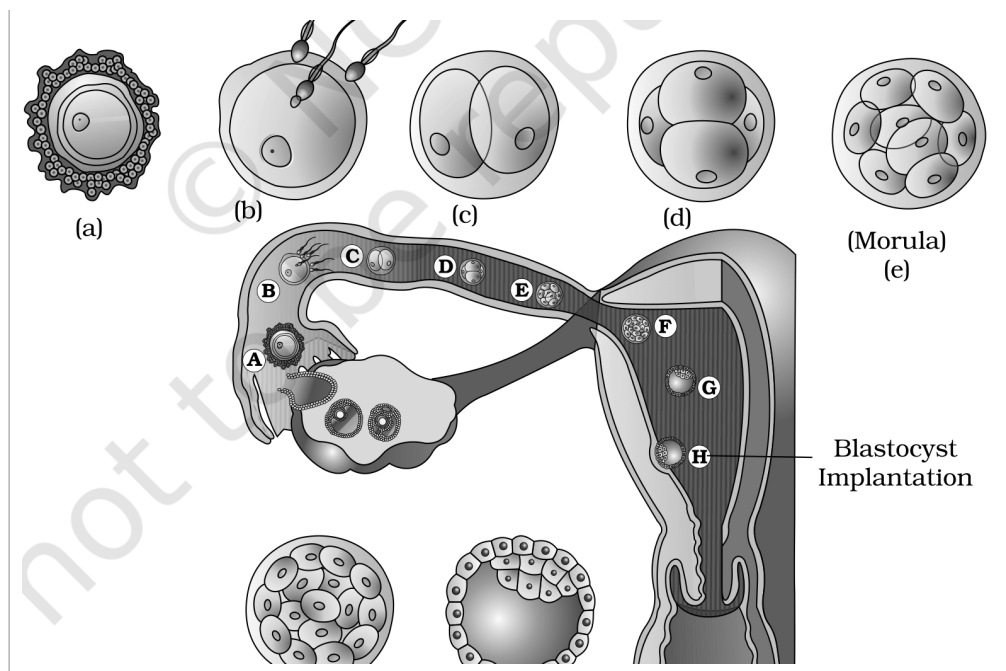


Fig. 2.11 - Transport of the ovum, fertilisation, and passage of the growing embryo through the fallopian tube. Labelled stages: morula, blastocyst and implantation.

2.6

Pregnancy and Embryonic Development

After implantation, **finger-like projections appear on the trophoblast called chorionic villi**, which are **surrounded by the uterine tissue and maternal blood**. The **chorionic villi and uterine tissue become interdigitated and jointly form a structural and functional unit between the developing embryo (foetus) and the maternal body called the placenta**.

The **placenta facilitates the supply of oxygen and nutrients to the embryo and the removal of carbon dioxide and excretory/waste materials produced by the embryo**. It is **connected to the embryo through an umbilical cord**, which helps in the transport of substances to and from the embryo.

The **placenta also acts as an endocrine tissue and produces hormones like human chorionic gonadotropin (hCG), human placental lactogen (hPL), estrogens, progestogens, etc.** In the **later phase of pregnancy**, a hormone called **relaxin** is also secreted by the ovary.

① **[NOTE]** *hCG, hPL and relaxin are produced in women only during pregnancy. In addition, during pregnancy the levels of other hormones like estrogens, progestogens, cortisol, prolactin, thyroxine, etc. are increased several-fold in the maternal blood. The increased production of these hormones is essential for supporting fetal growth, metabolic changes in the mother, and maintenance of pregnancy.*

Immediately after implantation, the inner cell mass (embryo) differentiates into an outer layer, the **ectoderm**, and an inner layer, the **endoderm**; a **mesoderm** soon appears between them. These **three (germ) layers give rise to all tissues (organs) in adults**. The inner cell mass contains certain cells called **stem cells** which have the **potency to give rise to all the tissues and organs**.

Human pregnancy lasts 9 months. The growth of the foetus follows a recognisable timeline:

Stage of pregnancy	Development observed
After 1 month	The embryo's heart is formed; the first sign of the growing foetus may be noticed by listening to the heart sound through a stethoscope.
End of 2 months	The foetus develops limbs and digits.
End of 12 weeks (first trimester)	Most major organ systems are formed; the limbs and external genital organs are well-developed.
Fifth month	The first movements of the foetus and the appearance of hair on the head are usually observed.
End of about 24 weeks (end of second trimester)	The body is covered with fine hair, the eye-lids separate, and eyelashes are formed.
End of 9 months	The foetus is fully developed and ready for delivery.

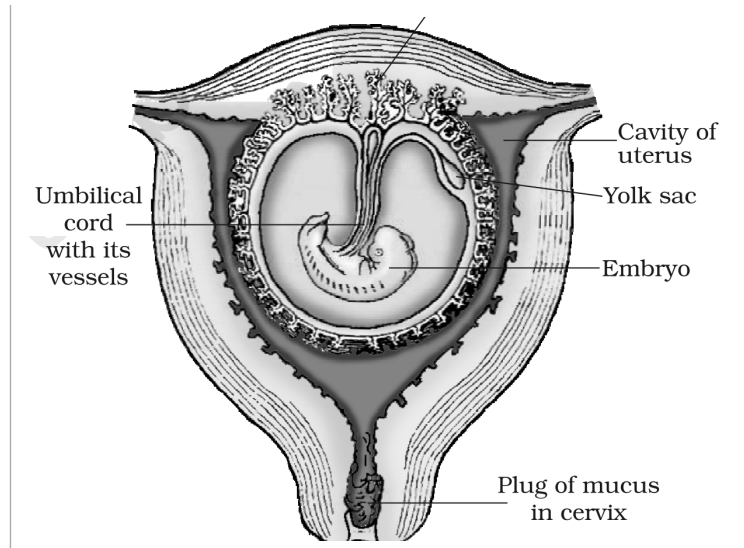


Fig. 2.12 - The human foetus within the uterus. Labelled parts: placental villi, umbilical cord, cavity of uterus, yolk sac, embryo and cervix.

2.7

Parturition and Lactation

The average duration of human pregnancy is about 9 months, which is called the **gestation period**. Vigorous contraction of the uterus at the end of pregnancy causes **expulsion/delivery of the foetus**. The process of delivery of the foetus (childbirth) is called **parturition**.

Parturition is induced by a complex neuroendocrine mechanism involving cortisol, estrogens and oxytocin. It proceeds as a self-reinforcing reflex:

- 1 Signals for parturition originate from the fully developed foetus and the placenta and induce mild uterine contractions called the **foetal ejection reflex**.
- 2 The foetal ejection reflex triggers the release of oxytocin from the maternal pituitary.
- 3 Oxytocin acts on the uterine muscle and causes stronger uterine contractions, which in turn stimulate further secretion of oxytocin.
- 4 This stimulatory reflex between uterine contraction and oxytocin secretion continues, resulting in stronger contractions and expulsion of the baby through the birth canal (parturition).

Soon after the infant is delivered, the placenta is also expelled out of the uterus.

The mammary glands undergo differentiation during pregnancy and start producing milk towards the end of pregnancy by the process called **lactation**. Lactation helps the mother in feeding the new-born. The milk produced during the initial few days of lactation is called **colostrum**, which contains several antibodies absolutely essential to develop resistance for the new-born babies. Hence **breast-feeding during the initial period of infant growth is recommended by doctors for bringing up a healthy baby**.

- Humans are **sexually reproducing and viviparous**. The male reproductive system is a **pair of testes**, the **male sex accessory ducts**, **accessory glands** and **external genitalia**.
- Each testis has **about 250 testicular lobules**, each with **one to three seminiferous tubules** lined by **spermatogonia (which undergo meiosis to form sperms)** and **Sertoli cells (which nourish the germ cells)**; the **Leydig cells outside the tubules secrete androgens**. The male external genitalia is the **penis**.
- The female reproductive system is a **pair of ovaries**, a **pair of oviducts**, a **uterus**, a **vagina**, **external genitalia** and a **pair of mammary glands**. The **ovaries produce the ovum and ovarian (steroid) hormones**, with **ovarian follicles at different stages embedded in the stroma**. The **uterus has three layers - perimetrium, myometrium and endometrium**, and the **mammary glands are a female secondary sexual characteristic**.
- **Spermatogenesis** yields sperms carried by the accessory ducts; a sperm has a **head, neck, middle piece and tail**. **Oogenesis** forms the mature female gamete. The **reproductive cycle of female primates is the menstrual cycle**, which starts only after puberty, releases **one ovum per cycle**, and is **controlled by pituitary and ovarian hormones**.
- After coitus, **sperms reach the ampulla, where a sperm fertilises the ovum to form a diploid zygote**; the **X- or Y-bearing sperm determines the sex**. The zygote forms a **blastocyst that is implanted in the uterus**, the **placenta connects foetus and mother**, and **gestation lasts about 9 months**.
- **Parturition is induced by a complex neuroendocrine mechanism involving cortisol, estrogens and oxytocin**; the **mammary glands secrete milk (lactation) after childbirth**, beginning with the antibody-rich **colostrum**.

Two exercise questions rely on ideas the chapter itself does not spell out. Both are answered here using only chapter-based reasoning.

- **Identical vs fraternal twins**. The chapter states that **one ovum is normally released per menstrual cycle** and that a **single sperm fertilises a single ovum** to form one zygote. **Identical twins** arise when **one fertilised ovum (one zygote) splits** to give two embryos - they share the same genetic make-up. **Fraternal (non-identical) twins** arise when **two separate ova are released and fertilised by two separate sperms**, giving two genetically different embryos.
- **Eggs released for a litter of six puppies**. Because each puppy develops from its own fertilised ovum (one zygote per offspring), a dog delivering a **litter of six puppies must have released at least six ova** in that cycle, each fertilised by a separate sperm.